INCH-POUND
MIL-C-24640/24B
31 July 1989
SUPERSEDING
MIL-C-0024640/24A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/24
10 August 1984

MILITARY SPECIFICATION SHEET

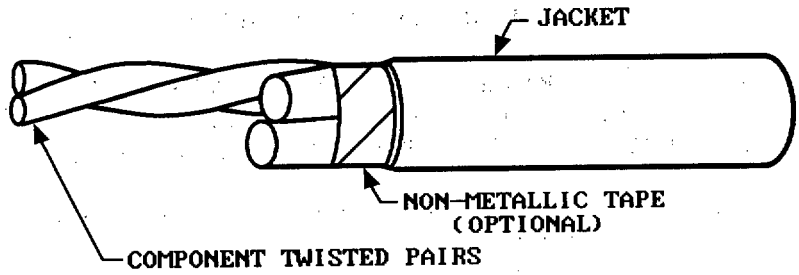
CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 22, TYPE TTXW (INCLUDING VARIATION TYPES TTXOW AND TTXWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

AMSC N/A
FSC 6145
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SH 12771

COMPONENT PAIRS:

(B) First - Tin coated copper conductor, AWG 22, 7 stranding in accordance with ASTM B 286. Nominal diameter of stranded conductor 0.031 inch.

Second - Coat wire with free-stripping waterblocking compound.

(B) Third - MICA-glass tape, insulation 0.0029 ± 0.0002 inch (50 percent overlap maximum).

Fourth - Opaque pigmented polyimide tape 0.001 inch (50 percent overlap minimum).

Fifth - Opaque pigmented polyimide tape 0.001 inch with 0.0001 inch fluorinated ethylene propylene (FEP) both sides (50 percent overlap minimum).

(B) Sixth - Modified aromatic polyimide resin coating 0.0005 inch minimum (0.001 inch nominal). Finished wire diameter 0.051 inch minimum and 0.055 inch maximum.

Seventh - One black coated wire twisted together with one white coated wire (except 1½ which shall be one black, one white, and one red wire twisted to form a triad). Twisted pair identification code applied by method 6.

FIGURE 1. Construction (watertight with circuit integrity).

(B) denotes changes.

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(B) TABLE I. Construction details-unarmored.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01UN	TTXW-1-1/2	1-1/2	0.028	0.181	0.195	26.0
-02UN	TTXW-3	3	.038	.285	.307	62.3
-03UN	TTXW-5	5	.038	.331	.357	91.0
-04UN	TTXW-10	10	.041	.456	.492	168.0
-05UN	TTXW-15	15	.044	.527	.569	231.0
-06UN	TTXW-20	20	.044	.577	.621	283.0
-07UN	TTXW-30	30	.044	.684	.738	407.0
-08UN	TTXW-40	40	.047	.790	.852	526.0

(B) TABLE II. Construction details-armored.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01AN	TTXWA-1-1/2	1-1/2	0.028	0.231	0.245	41.0
-02AN	TTXWA-3	3	.038	.335	.357	84.3
-03AN	TTXWA-5	5	.038	.381	.407	119.4
-04AN	TTXWA-10	10	.041	.506	.542	209.3
-05AN	TTXWA-15	15	.044	.577	.619	269.6
-06AN	TTXWA-20	20	.044	.627	.671	328.2
-07AN	TTXWA-30	30	.044	.734	.788	463.2
-08AN	TTXWA-40	40	.047	.840	.902	592.6

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(B) TABLE III. Construction details-overall shield.

PIN M24640/24	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01UN	TTXOW-1-1/2	1-1/2	0.127	0.031	0.253	0.273	67.9
-02UN	TTXOW-3	3	.216	.038	.333	.359	108.0
-03UN	TTXOW-5	5	.263	.038	.376	.406	134.0
-04UN	TTXOW-15	15	.449	.044	.556	.600	299.0
-05UN	TTXOW-20	20	.510	.044	.614	.662	361.0
-06UN	TTXOW-30	30	.610	.047	.717	.772	480.0
-07UN	TTXOW-40	40	.720	.047	.823	.887	631.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Finished wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature.
Voltage rating: 600 volts root mean square (rms) at sea level
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F); no sticking of conductors.
Color: As specified in MIL-C-24640.
Flammability: 3 seconds (maximum) after-flame.
3.0 inches (maximum) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms-1,000 feet, minimum insulation resistance after humidity exposure.
(B) Identification of product: Twisted pair identification code applied by method 6.
Impulse dielectric test: 100 percent test; impulse voltage at 8 kV.
Insulation resistance: 2,500 megohms-1,000 feet (minimum).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch (maximum) at 180°C.
Surface resistance: 5 megohms-inches (minimum), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Change in measurement, 0.031 inch (maximum).
Wet dielectric test: 2,500 volts rms.

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REQUIREMENTS: - Continued

When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) Braided metal armor shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(B) (a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 90 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(B) Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.54
(B) Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (TTXOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	2500
Conductor and shield continuity (type TTXOW only)	No failure
Jacket flaws (type TTXOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	No failure

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EXAMINATION AND TESTS: - Continued

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type TTXWA only) - conformance to material, construction and coverage	No failure
Shield (type TTXOW only) - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance (type TTXOW only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: - Continued

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	No failure
Finished wire	No failure

Preparing activity:
Navy - SH
(Project 6145-1004-24)

Custodians:
Army - MI
Navy - SH

Review activities:
Navy - EC
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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